Module 02: Agents — The Self-Model, Interoception, and Ego Boundaries

Learning Objectives

1. Describe how the brain constructs a **self-model** through interoceptive inference — the perception of internal bodily states.
2. Explain the role of the insular cortex and anterior cingulate cortex in maintaining the agent's sense of self.
3. Analyze clinical conditions (depersonalization, rubber hand illusion, body integrity disorder) as disruptions of the self-model.
4. Connect the Active Inference self-model to the **4E cognition** framework — embodied, embedded, enactive, and extended cognition.

Introduction

What makes you *you*? Cognitive science reveals that the sense of self is not a given but a construction — an ongoing inference about the boundaries, states, and capacities of your own body. Active Inference formalizes this: the **self-model** is the component of the generative model that represents the agent's own Markov Blanket. This module examines the neural substrates, clinical disruptions, and philosophical implications of this self-model.

Key Concepts

1. Interoception as Self-Inference

Interoception — the perception of internal bodily states (heartbeat, breathing, gut sensations, temperature) — is the sensory channel through which the brain maintains its model of the body. Anil Seth (2013) argues that the "self" is fundamentally an interoceptive prediction: the brain's best guess about the causes of its internal sensory signals.

The **insular cortex** (particularly the anterior insula) is the primary cortical area for interoceptive processing. It receives input from the vagus nerve, sympathetic afferents, and chemoreceptors, and integrates these signals into a coherent representation of bodily state. In Active Inference terms, the anterior insula maintains the self-component of the generative model.

Craig (2009) showed that interoceptive accuracy correlates with emotional awareness: individuals who can more accurately count their heartbeats also show greater emotional granularity and metacognitive insight, consistent with a higher-precision self-model.

2. Body Ownership and the Rubber Hand Illusion

The **rubber hand illusion** (Botvinick & Cohen, 1998) demonstrates that body ownership is inferential. When a participant sees a rubber hand being stroked while their own hidden hand is stroked simultaneously, the brain integrates the visual and tactile signals and "reassigns" ownership to the rubber hand.

In Active Inference terms, the generative model updates its representation of the body boundary (the Markov Blanket) based on the best explanation of the multisensory evidence. The rubber hand is incorporated into the self-model because this reduces overall prediction error across modalities.

Crucially, this update also affects **proprioception**: the felt position of the real hand drifts toward the rubber hand (**proprioceptive drift**), demonstrating that the self-model is not just visual but multimodal.

3. Clinical Disruptions of the Self-Model

- **Depersonalization disorder**: Patients report feeling detached from their own body and mental processes. Active Inference interpretation: low precision on interoceptive prediction errors — the self-model becomes uncertain and attenuated.
- **Body integrity identity disorder (BIID)**: Patients feel that a healthy limb does not belong to them and may seek amputation. Interpretation: the generative model's body representation excludes the limb, creating persistent prediction error between the model and the actual body.

- **Autism spectrum**: Some individuals show heightened interoceptive awareness combined with difficulty integrating interoceptive signals into a coherent self-model, contributing to sensory overwhelm and alexithymia (difficulty identifying emotions).
- **Phantom limb pain**: After amputation, the generative model continues to predict sensory and proprioceptive input from the missing limb, producing vivid phantom sensations and sometimes pain — a dramatic case of model-world mismatch.

4. The 4E Cognition Framework

The self-model in Active Inference connects to a broader movement in cognitive science known as **4E cognition** (Newen, de Bruin, & Gallagher, 2018):

- **Embodied**: Cognition depends on the physical body. The generative model necessarily includes a body model — without one, the agent cannot distinguish self-generated from externally-generated sensory signals.
- **Embedded**: The agent's model is structured by its ecological niche. A bat's self-model includes echolocation; a human's includes speech production.
- **Enactive**: Perception is not passive observation but active engagement with the world (Varela, Thompson, & Rosch, 1991). Active Inference formalizes this through the action-perception loop: the agent acts to fulfill its predictions.
- **Extended**: Tools and technologies can become part of the self-model. A blind person's cane, a surgeon's instruments, or a smartphone may function as extensions of the Markov Blanket when they are reliably incorporated into sensorimotor predictions.

5. Vestibular Self and Spatial Agency

The **vestibular system** provides a foundational self-signal: where am I in space? Vestibular input from the semicircular canals and otolith organs anchors the agent's spatial self-model. Lopez and Blanke (2011) showed that vestibular stimulation can induce out-of-body experiences, suggesting that the spatial self-model is actively inferred and can be disrupted.

This connects to the sense of **agency** — the feeling that "I am the one doing this." The comparator model (Blakemore, Wolpert, & Frith, 2002) proposes that agency arises when the predicted sensory consequences of an action match the actual consequences. Active Inference generalizes this: agency is the precision-weighted alignment between predicted and observed outcomes of self-initiated actions.

Conclusion

The self is not a homunculus but an inference — a continuously updated model of the agent's own body, boundaries, and capacities. The 4E cognition framework situates Active Inference within the broader tradition of embodied, embedded, enactive, and extended mind. Understanding the neural and clinical dimensions of this self-model is essential for connecting Active Inference to psychopathology, embodiment, and subjective experience. Module 03 examines how the self-model interfaces with the external world through perception.

Further Reading

- Seth, A. K. (2013). Interoceptive inference, emotion, and the embodied self. *Trends in Cognitive Sciences*, 17(11), 565-573.
- Botvinick, M. & Cohen, J. (1998). Rubber hands 'feel' touch that eyes see. *Nature*, 391, 756.
- Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind*. MIT Press.
- Blakemore, S. J., Wolpert, D. M., & Frith, C. D. (2002). Abnormalities in the awareness of action. *Trends in Cognitive Sciences*, 6(6), 237-242.